SUPPLY CHAIN DELIVERY PERFORMANCE REPORT

Global E-Commerce Fulfilment Analytics

January 2015 — January 2018 | 172,765 Orders

Table of Contents

1. Executive Summary

This report presents a comprehensive analysis of the delivery operations of a global e-commerce company managing end-to-end order fulfilment across multiple regions. The analysis covers 172,765 orders spanning January 2015 through January 2018, focusing on identifying root causes of chronic late deliveries, quantifying their financial impact, and establishing a data-driven framework for improvement.

The headline finding is stark: 54.71% of all orders are delivered late, costing the business $2.1M in eroded profitability on delayed orders alone. While total profit across fulfilled orders reached $7.5M, the persistent late-delivery problem represents both a significant financial drain and a major customer experience risk. A Random Forest predictive model achieved 74% accuracy in identifying at-risk orders before they become late.

Key Finding	Metric
Overall Late Delivery Rate	**54.71%**
On-Time Delivery Rate	**42.29%**
Total Orders Analyzed	**172,765**
Total Profit (Profitable Orders)	**$7.5M**
Profit at Risk (Delayed Orders)	**$2.1M**
Predictive Model Accuracy (RF)	**74%**

2. Business Context & Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to buyers at the point of purchase.

Analytical Objectives

- Understand the current state of delivery performance across all dimensions (region, mode, time, segment)
- Quantify the financial impact of delays on order profitability
- Identify primary operational bottlenecks driving late deliveries
- Build a predictive model to flag high-risk orders before they are shipped
- Deliver actionable recommendations to improve on-time delivery and profitability

3. Key Performance Indicators

The following KPIs are computed directly from the cleaned dataset to establish a performance baseline.

172,765 Total Orders	**94,523** Late Deliveries	**54.71%** Late Delivery Rate	**45.29%** On-Time Delivery Rate
$7.5M Total Profit	**$2.1M** Profit at Risk	**3 Days** 90th Percentile Delay	**$22.03** Avg Order Profit

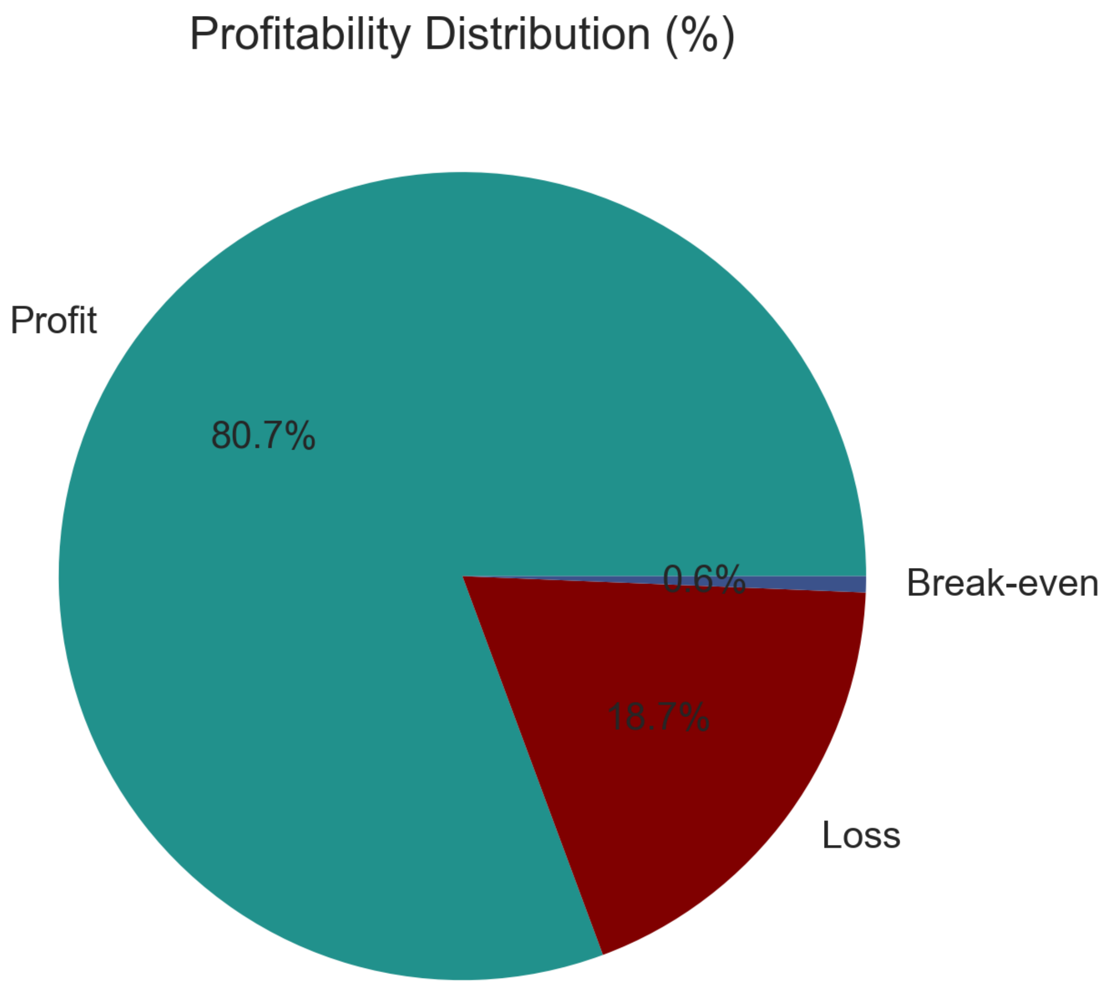
KPI Interpretation

Indicator	Interpretation
Late Delivery Rate: 54.71%	More than half of all orders arrive late. This is not an edge case — it is the default experience for the majority of customers.
$2.1M Profit at Risk	Orders that experienced delays collectively hold $2.1M in profit under constant pressure from further operational deterioration.
90th Percentile Delay = 3 Days	Even the most extreme cases contain only 3 days of lateness, suggesting the problem is systematic and process-driven rather than catastrophic.

4. Profitability Analysis

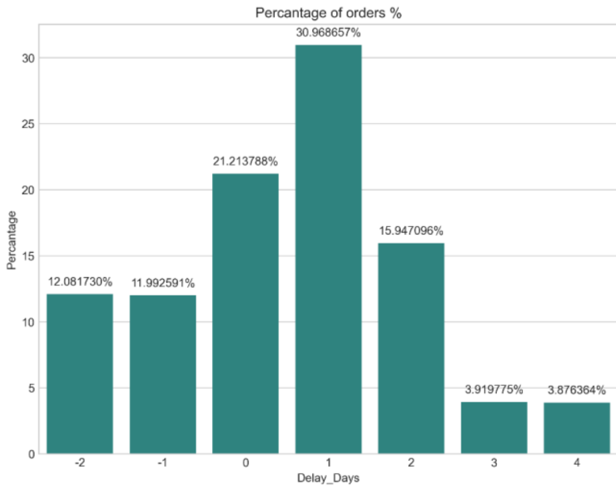
4.1 Profitability Distribution

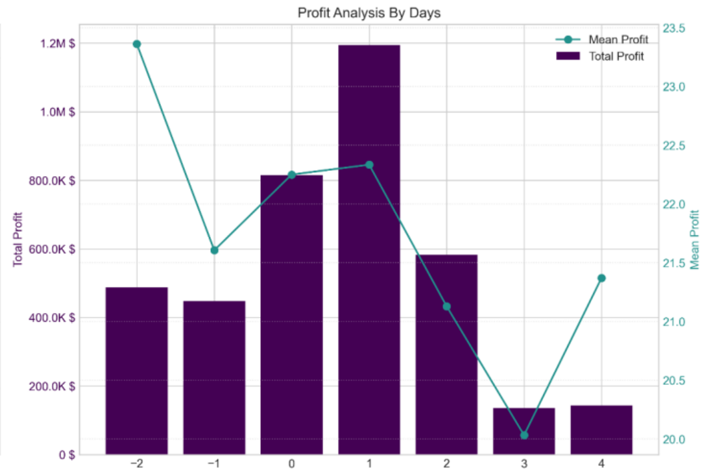
Order-level profitability was classified into three tiers based on profit per order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.



4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly 1 day late — the single largest cohort. Combined, orders delayed 1–4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.

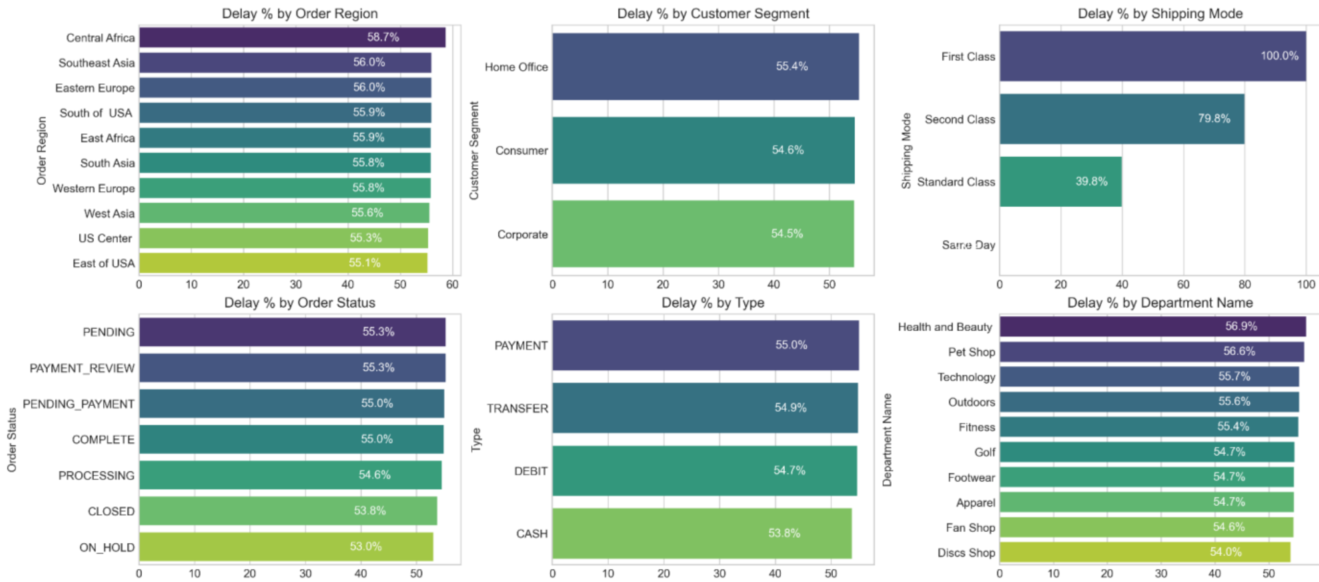




> **Critical Insight:** Mean profit per order remains stable at approximately $21–23 across all delay levels. The profitability problem is therefore driven by the volume of delayed orders, not by individual order economics — making systematic throughput improvement the highest-leverage intervention.

5. Bottleneck Detection

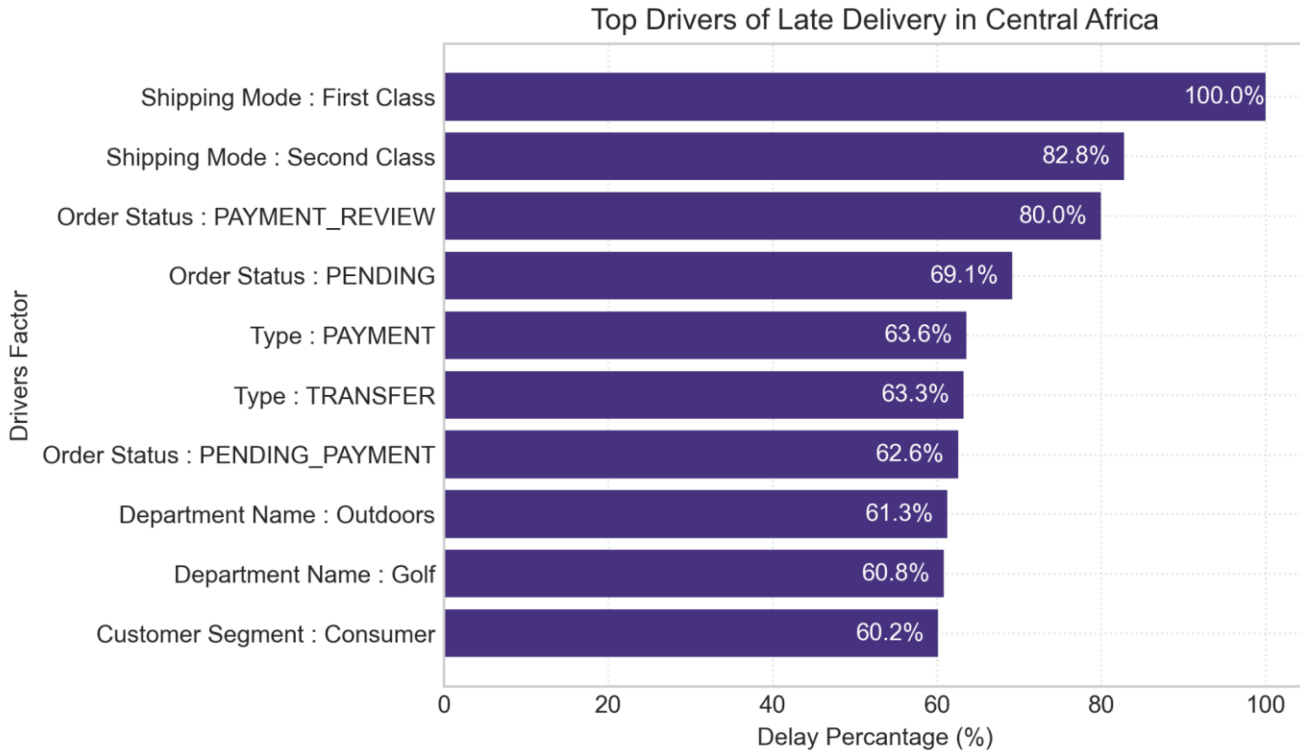
Delay percentage was computed across six key operational dimensions. The charts below reveal where fulfilment breaks down most severely.



Finding	Detail
Shipping Mode: #1 Lever	First Class: 100% delay rate. Second Class: 79.8%. Standard Class: 39.8%. Same Day: 0%. The mode assignment logic is the most impactful single variable to address.
Regional Spread Narrow (55–59%)	Central Africa leads at 58.7%, but all regions cluster between 55–59%. This rules out a localised logistics failure and confirms a company-wide systemic issue.
Customer Segment Nearly Identical (54.5–55.4%)	No segment receives preferential service. All customers — Consumer, Corporate, and Home Office — experience the same broken delivery promise.
Health & Beauty (56.9%) and Pet Shop (56.9%) Lead by Department	These departments marginally outpace others in delay rate, warranting an inventory and carrier audit for these categories.

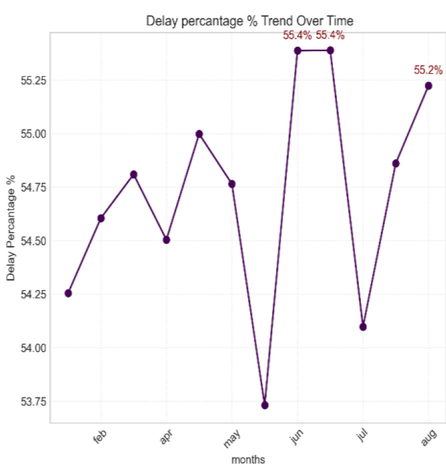
6. Root Cause Analysis

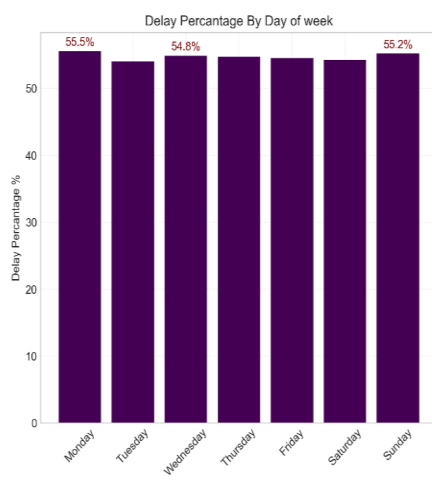
Central Africa was selected for a deep-dive root cause analysis as the highest-delay region (58.7%). The chart below ranks the top driver factors by delay percentage within this region.

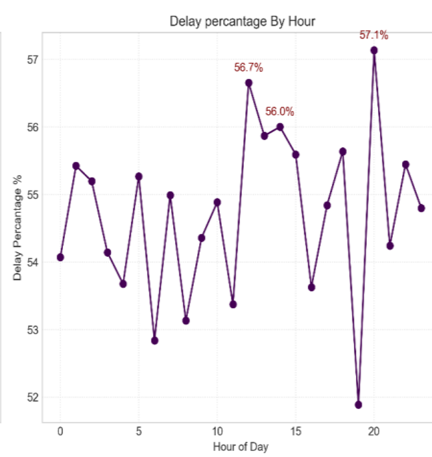


7. Time-Based Delay Patterns

Three temporal dimensions were analysed: month of year, day of week, and hour of day. While delay rates are relatively stable across all dimensions (53–57%), specific peaks highlight opportunities for targeted capacity planning.







Pattern	Insight
Peak Delay Months: August & September (55.4%) and December (55.2%)	August and September are the highest months at 55.4%, with December close behind. These reflect mid-year promotions and the Q4 holiday surge overwhelming fulfilment capacity. July represents a notable low point (~53.75%), confirming that seasonal variation is plannable.
Day-of-Week Variance is Minimal	Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than 1 percentage point. Day of week is not a meaningful lever for operational intervention.
Intraday Peaks: Hour 21 (57.1%), Hour 11 (56.7%), Hour 12 (56.0%)	Late-evening orders (hour 21) have the highest delay rate at 57.1%, likely reflecting processing cutoff windows. Midday peaks suggest bottlenecks during peak volume hours.

8. Machine Learning Model Results

A supervised classification model was built to predict whether an individual order will be delivered late (Late_delivery_risk = 1). The pipeline includes frequency encoding of categorical variables, a stratified train/test split, SMOTE oversampling to address class imbalance, and Random Forest classification.

Class Imbalance Handling (SMOTE)

Split	Class 0 (On-Time)	Class 1 (Late)
Before SMOTE	59,030	79,182
After SMOTE	79,182	79,182

Random Forest Classifier — Test Set Performance (34,553 Records)

Metric	Class 0 (On-Time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1-Score	0.70	0.77	0.74
Accuracy	—	—	0.74
Support	14,758	19,795	34,553

Model Interpretation

Finding	Interpretation
74% Overall Accuracy	The model correctly classifies nearly 3 in 4 orders — a strong foundation for production deployment as an order-level late-delivery alert system.
Higher Precision on Late Orders (0.78)	When the model predicts an order will be late, it is correct 78% of the time — sufficient to trigger targeted interventions without generating excessive false alarms.
Recall for Late Orders: 0.75	The model captures 75% of actual late orders. The 25% miss rate is acceptable for a first-generation alert system with clear improvement potential.
Class 0 Harder to Predict (Precision 0.68)	On-time deliveries share characteristics with late ones, given the systematic nature of delays, making them genuinely difficult to distinguish.

9. Strategic Recommendations

The following eight recommendations are prioritised by impact and implementation urgency. Colour coding reflects strategic priority level.

Priority	Recommendation	Target KPI Impact
CRITICAL	**1. Immediately Audit First & Second Class Shipping Capacity** First Class has a 100% delay rate; Second Class: 79.8%. These modes operate in complete contradiction to their brand promise. Conduct an immediate carrier SLA audit. Until resolved, consider suspending First Class or repricing to reflect actual performance.	**Reduce overall late rate from 54.71% to <45%**
HIGH	**2. Deploy the Predictive Alert System** The Random Forest model (74% accuracy, 78% precision on late orders) is ready for production piloting. Integrate it into the order management system to flag high-risk orders at confirmation and trigger proactive customer communication and priority handling queues.	**Intercept 75% of at-risk orders pre-shipment**
HIGH	**3. Resolve Payment Processing Bottlenecks** Orders in payment review (55.3% overall; 80% in Central Africa) and pending payment (55.0%) have significantly elevated delay rates. Introduce automated escalation for stalled payment reviews exceeding 2 hours.	**Reduce payment-related delays by ~25%**
MEDIUM	**4. Develop Seasonal Surge Capacity Plans** August–October (55.4%) and December (55.2%) are peak delay months. Build pre-season capacity plans including temporary logistics partnerships, inventory pre-positioning, and adjusted delivery promise windows.	**Reduce seasonal peak delay by ~5%**
MEDIUM	**5. Default to Standard Class for Eligible Orders**	**Improve mode-level delay rates by 15–40%**

	Standard Class achieves a 39.8% delay rate — far superior to First and Second Class. Re-evaluate the shipping mode assignment logic to default eligible orders to Standard Class, particularly in regions and departments with poor premium mode performance.	
MEDIUM	**6. Investigate High-Delay Departments in Africa** Outdoors (61.3%) and Golf (60.8%) in Central Africa show significantly elevated rates. Conduct a warehouse-level audit of stock availability, pick-and-pack times, and carrier assignment for these product categories.	**Reduce Africa dept delays to <55%**
LOW	**7. Reduce Loss-Making Order Share (18.7%)** The 18.7% loss-making order share warrants a separate pricing and discount review. Combine with delay analysis to identify whether discounted orders are disproportionately late, creating a compound margin problem.	**Reduce loss-making share to <12%**
LOW	**8. Continuously Retrain the Predictive Model** Retrain the model quarterly. Explore adding carrier-level data, weather events, and warehouse utilisation features. Target a recall improvement to above 80% on late orders within 6 months.	**Improve model recall from 75% to 80%+**

COLOUR LEGEND			
CRITICAL — Red	**HIGH — Amber**	**MEDIUM — Blue**	**LOW — Green**

10. Conclusion

This analysis has surfaced a clear and urgent picture: a global e-commerce operation where the majority of orders (54.71%) fail to meet their promised delivery windows, costing $2.1M in at-risk profit and undermining customer trust at scale.

The root causes are identifiable and addressable. Shipping mode misconfiguration (First Class at 100% late, Second Class at 79.8%) is the dominant operational failure. Payment processing friction creates a secondary bottleneck. Seasonal volume spikes are not adequately planned for. Geographic disparities, while present, are secondary to the systemic company-wide pattern.

A Random Forest predictive model trained on readily available order features already achieves 74% accuracy in identifying at-risk orders. This is a deployable tool, not a future aspiration.

Priority Area	Current State	Target State
Late Delivery Rate	**54.71%**	**<30% within 12 months**
First Class On-Time Rate	**0%**	**>80%**
Second Class On-Time Rate	**20.2%**	**>60%**

Predictive Model Accuracy	**74%**	**>82% (enhanced model)**
Loss-Making Orders	**18.7%**	**<12%**
Profit at Risk	**$2.1M**	**Reduce by 40%**